CaffeinatedData

Tableu SQL Project

The *Churn* dataset will be utilized, along with an additional complimenting data set '**CGB - Consumer Complaints Data'**. From opendata.fcc.gov

https://opendata.fcc.gov/Consumer/CGB-Consumer-Complaints-Data/3xyp-aqkj/about_data

'Individual informal consumer complaint data detailing complaints filed with the Consumer Help Center beginning October 31, 2014. This data represents information selected by the consumer. The FCC does not verify the facts alleged in these complaints.'

Last Updated: March 7, 2026

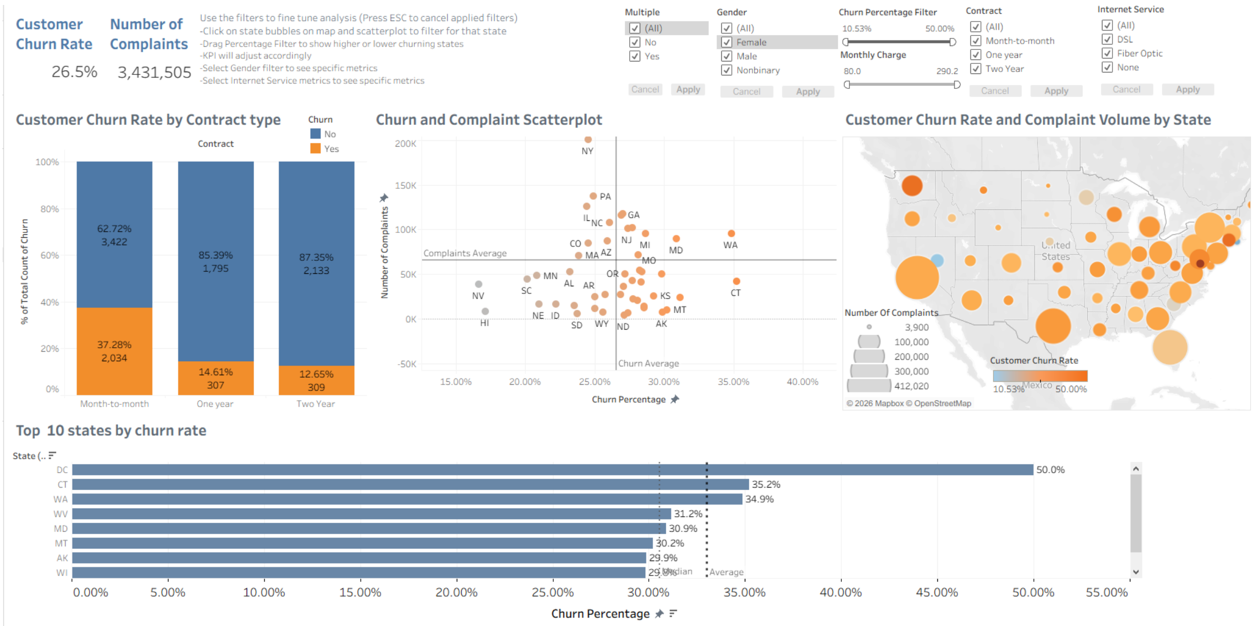
Data Provided By: Federal Communications Commission: Consumer Inquiries and Complaints Division

The data has been filtered to keep relevant columns and aggregated to keep the file small and remove noise while staying workable as larger files can make the program slow and unresponsive.

The file is essentially a summary of telecom related consumer complaints in the U.S.

It involves complaints like billing disputes, telemarketing, service availability, internet speed, signal problems, unwanted calls. The data will relate to the churn data set by the '**State'** column.

This will provide insight into how Federal level run surveys on complaints involving telecom companies which can affect churn.



Dashboard Purpose

The purpose of the dashboard is to analyze patterns in customer churn and identify factors that may influence customer retention. The variable used in the dashboard align with the data dictionary of the customer churn data set, which includes attributes such as churn status, contract type, gender, monthly charge and internet service. These variables help describe the characteristics of customers who remain with the service compared to those who leave. The dashboard uses these variables to visualize churn trends across different categories and geographic locations. By composing the variables into visual representations, such as bar charts, maps and scatterplot, it becomes easier to notice a relationship and understand the data that otherwise wouldn't have been easily discovered through a normal table.

The additional dataset containing consumer complaints of telecom companies enhances the sight of the customer churn dataset by providing a broader perspective on customer dissatisfaction. While the churn dataset identifies customers who left the service, the complaint dataset provides information about the number of complaints reported in each state. Combining these two datasets allows the analysis to explore whether areas with higher complaint volumes also experience higher churn rates. This integration helps provide a more easily consumable view of customer behavior and may geographically indicate regions where service improvement is required.

Leadership Usage:

The two key data representations used in the dashboard are the scatterplot and the contract churn bar chart. The scatterplot visualizes the relationship between customer churn rates and the number of complaints reported in each state. Shareholder leaders can use this chart to evaluate whether customer dissatisfaction is associated with increased churn levels. The contract churn bar chart compares churn rates across different contract types. This representation allows executives to quickly identify that customers with month-to-month contracts experience higher churn rates compared to customers with longer term contracts. Such information can be utilized to create a customer retention plan.

Interactive Controls

The dashboard includes several interactive controls that allow users to modify and alter the presentation of the data. One interactive control is the map filter. Users could click on the state bubbles in the map to filter for that specific state in which it would alter the KPI metrics and bar chart to showcase metrics for that choice. Another interactive control is the Monthy Charge slider, which allows the user to adjust the monthly charge and see how it affects the charts, in turn displaying different churn rates on the map, the bar chart and the scatterplot.

Colorblind accessibility:

The dashboard was designed to be accessible for individuals with color blindness by using a friendly color palette. For categorical fields, Tableau has an integrated palette that acknowledges the vision problems that colorblind users face. As for continuous fields, a blue-orange gradient color scheme is adopted to distinguish between different levels of churn to avoid the red-green color combinations that may be difficult for some users to distinguish. In addition to color, visual elements such as circle size and labels were used to represent information. Using multiple visual encodings ensures that users can interpret the data even if they have difficulty distinguishing certain colors.

How data supports the story:

Two visualizations that support the story of the analysis are the geographic map, and the top ten states bar chart. The map visualizes churn rates and complaint volume by state, allowing users to identify geographic patterns in customer dissatisfaction and churn behavior. The bar chart displays the top ten states by churn rates highly which regions experience the highest levels of customer churn. Together, these visualizations help illustrate the broader narrative that churn rates may vary by location and may be influenced by other factors.

Audience Analysis:

The intention to make this dashboard tailored to shareholders is to make the visualizations easy to digest. As such, emphasizing key metrics and easily absorbable information was key to tailoring this for the audience. The dashboard prominently displays KPI indicators, a scatterplot to showcase outlying states, while also showing states that are clustered to identify regions where finding a solution to the churn rate is much more achievable. This approach ensures that decision makers can quickly understand the key insights from the analysis.

Universal audience:

The dashboard was designed to support a broad range of users. Clear labeling, readable fonts, and simple visual layout with headers to signify the information the chart will provide.

Interactive filters allow the user to customize the dashboard based on their interest and analytical needs. While the charts can act as a filter in order to fine tune the modification to specify region-based data. This is while the KPI metrics and bar chart on contract types are adjusted to match that specific region. The scatterplot helps look for target regions that need more investigation, working back and forth with the map as they can help the user find a specific state.

Storytelling:

Two elements of effective storytelling are implemented in the dashboard, dynamic visualization and contextual narrative. Dynamic data visualization is used throughout the dashboard to communicate how metrics change when different filters are applied. It makes complex data easier to interpret. These visualizations help the audience see clear differences in the data. Context is provided throughout the organization of the dashboard and the use of KPIs, which summarizes overall churn and complaint levels before users explore deeper. Because the map, charts, and plots are integrated into a system that responds to changes.